

CRISTOFF STAN

Senior Software Engineer

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PROFESSIONAL SUMMARY

Machine Learning Tech Lead with 8+ years of experience in architecting AI-driven solutions across cloud-native environments, specializing in LLMs, MLOps, and NLP pipelines. Proven track record of leading cross-functional engineering teams in building scalable ML infrastructures using AWS (SageMaker, Lambda, S3, Bedrock) and deploying state-of-the-art models (OpenAI, Hugging Face) for real-time inference. Strong technical leadership in applying deep learning, reinforcement learning, and classical ML to solve business challenges in Healthcare, Retail, and Internet verticals. Passionate about mentoring engineers, enforcing engineering excellence, and continuously advancing ML innovation.

TECHNICAL SKILLS

- **Languages:** Python, TypeScript, JavaScript, SQL, Bash
- **AI/ML Frameworks:** PyTorch, TensorFlow, Hugging Face Transformers, OpenAI API, Scikit-learn, spaCy
- **Cloud & DevOps:** AWS (SageMaker, Lambda, Bedrock, ECS, ECR, S3, IAM, CloudWatch), Docker, GitHub Actions, Terraform
- **MLOps/LLMOps:** MLflow, KServe, Seldon Core, DVC, Airflow, Kubeflow, CI/CD for ML, Prompt engineering
- **Frontend:** React.js, Next.js, Angular, HTML5, CSS3
- **Databases:** PostgreSQL, MongoDB, DynamoDB, Redis
- **Concepts:** LLM fine-tuning, RLHF, prompt chaining, distributed training, real-time ML inference, NLP, CV, classical ML, vector databases (FAISS, Pinecone)

WORK EXPERIENCE

Senior Software Engineer

Sensidev

September 2021 – December 2024

Timișoara, Romania (Remote)

- Spearheaded the design and deployment of transformer-based LLM services for customer support automation, improving ticket classification accuracy by over 27% across multilingual datasets.
- Led a cross-functional squad of 8 engineers to deliver end-to-end ML pipelines using SageMaker, DVC, and Airflow, driving production-grade deployment of NLP and CV models.
- Implemented retrieval-augmented generation (RAG) using OpenAI and vector DBs to power a semantic search system across 4TB of internal documents with sub-second latency.
- Refactored model serving architecture by containerizing models via Docker and deploying through AWS ECS/ECR, reducing latency by 40% and cost by 22%.

- Developed prompt chaining workflows and tools for internal LLM prototyping, enabling rapid experimentation with few-shot and zero-shot tasks for product teams.
- Built a CI/CD pipeline for ML with GitHub Actions and Terraform to automate training, testing, and release of ML artifacts with rollback and monitoring integrations.
- Advocated and enforced engineering best practices through code reviews, performance benchmarks, and sprint planning, mentoring 4 junior ML engineers.
- Acted as a technical liaison between product and engineering leadership, aligning roadmap features with data insights and ML strategy.

Software Engineer

Syscom Digital

October 2019 – August 2021

Bucharest, Romania (Remote)

- Built ML-powered personalization modules for real-time marketing campaigns using TensorFlow, achieving a 17% lift in engagement through behavior-based targeting.
- Created a modular MLOps stack on AWS using SageMaker, Lambda, and S3 to orchestrate batch prediction jobs, version control, and automated scaling of endpoints.
- Integrated Hugging Face models for sentiment analysis and text summarization into CRM tools, reducing manual review overhead for customer success teams.
- Optimized Python data pipelines using pandas, NumPy, and Spark for ETL processing of 50M+ records per day across campaign analytics and A/B test metrics.
- Conducted in-depth model explainability studies using SHAP and LIME to support decision-making and regulatory reporting in GDPR-compliant systems.
- Introduced a monitoring dashboard via CloudWatch and Prometheus to detect ML model drift and automate retraining triggers based on confidence scores.
- Participated in strategic planning for AI roadmap adoption, identifying business cases for computer vision in customer recognition and QR engagement.
- Mentored junior data engineers on scalable coding practices and ML model evaluation techniques.

Software Developer

Soft Build

June 2016 – July 2019

Timișoara, Romania (Remote)

- Collaborated with data scientists to integrate ML models into web-based logistics apps, enabling predictive ETA calculation with 94% accuracy using historical route data.
- Reengineered backend APIs in Python and Node.js to interface with machine learning inference layers hosted on AWS Lambda and API Gateway.
- Built data preprocessing modules for large CSV and Parquet files, using PySpark and Dask, significantly accelerating model training pipelines.
- Developed internal tools to automate the labeling of image data using computer vision models trained on TensorFlow and augmented with Albumentations.
- Created comprehensive unit and integration test suites for ML microservices with PyTest, ensuring code reliability and performance in production.
- Integrated user behavior analytics for frontend applications to feed back into training datasets, improving model relevance over time.
- Implemented an event-driven architecture using AWS SNS/SQS for asynchronous processing of image and text data streams.
- Participated in internal R&D on federated learning architectures and their implications for GDPR and privacy-centric deployment.

Software Developer

Decima Digital

November 2015 – April 2016

Tallinn, Estonia

- Assisted in refactoring monolithic applications into microservices to support containerized ML workloads on Docker and Kubernetes.
- Built RESTful APIs to deliver prediction services from scikit-learn and XGBoost models exposed via Flask and deployed on AWS EC2 instances.
- Enhanced caching and data access logic for ML-powered dashboards, enabling real-time insights using Redis and PostgreSQL views.
- Supported exploratory data analysis on marketing datasets to inform business analysts of performance drivers using matplotlib and seaborn.
- Contributed to documentation of ML pipelines, including data flow diagrams, feature descriptions, and infrastructure setup.
- Integrated log-based user tracking mechanisms to generate feature sets for recommender systems across web and mobile applications.
- Participated in sprint cycles, performing peer code reviews and tracking bug reports related to model performance and deployment issues.
- Collaborated on security review and IAM setup to comply with AWS best practices around ML endpoints and data access layers.

Software Developer Intern

Borna Technology

September 2014 – August 2015

Tallinn, Estonia (On-Site)

- Developed initial ML prototypes for text classification of customer queries using Naive Bayes and SVMs, achieving 82% macro F1 score.
- Built lightweight dashboards for visualizing sentiment scores and keyword trends using Chart.js, D3.js, and REST APIs.
- Participated in data collection and annotation efforts for training corpora, contributing over 10,000 labeled examples for supervised models.
- Created pipeline scripts to parse and preprocess email and chat logs, including tokenization, stop-word removal, and lemmatization.
- Designed and tested SQL queries for extracting insights from CRM systems to improve training dataset quality and coverage.
- Contributed to a proof-of-concept chatbot using rule-based NLP approaches and early RNN models with Keras.
- Collaborated closely with QA and engineering to evaluate model accuracy and system stability under live usage.
- Documented experiments and model performance benchmarks to support transition into full-time development.

EDUCATION

Tallinn University

Bachelor's Degree in Computer Science,

2010 – 2014

CERTIFICATIONS

- ❖ AWS Certified Machine Learning – Specialty
 - ❖ DeepLearning.AI Generative AI with LLMs Specialization
 - ❖ Google Cloud MLOps Engineering Specialization
 - ❖ Hugging Face Transformers Course
 - ❖ TensorFlow Developer Certificate
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PROJECTS

Real-Time LLM Document Assistant

- Designed and deployed a scalable LLM document Q&A system using OpenAI and Pinecone, enabling instant answers across 1M+ company records with AWS Bedrock and Lambda.

MLOps Platform Automation

- Developed a reusable MLOps boilerplate using SageMaker Pipelines, MLflow, DVC, and Terraform, reducing time-to-deploy for new models by 60% across 3 internal teams.

AI-Powered Personalization Engine

- Engineered a user segmentation and recommendation system for eCommerce using PyTorch and XGBoost, driving a 24% increase in CTR and 15% uplift in revenue per user.